

Major Equipment Schedule and Technical Particulars

This document summarizes the principal equipment families used in the Prairie Horizon Data Campus interconnection planning basis. Values are provided to support utility review of configuration, rating, and general design intent. Final manufacturer submittals, dimensions, short-circuit data, relay complements, and detailed test values will be issued during procurement and detailed design.

1. Main Power Transformers

Three main power transformers are assumed, each rated 138/13.8 kV with OA/FA/FA ratings of 100/133/167 MVA. The planning basis assumes oil-filled outdoor units with high-side load tap changers, transformer differential protection, sudden-pressure monitoring, oil-containment basins, and fire separation between adjacent units.

Planning item	Assumed value
Voltage ratio	138 kV high side / 13.8 kV low side
Cooling class	OA/FA/FA
Ratings	100 / 133 / 167 MVA
Impedance	Approximately 9.5% on 133 MVA base for planning
Protection basis	87T, 63 sudden pressure, 24 overexcitation, temperature / thermal supervision
Civil basis	Individual containment, transformer foundations, and firewalls between adjacent units

2. 13.8 kV Main Switchgear

The campus primary distribution basis uses Siemens NXAIR M medium-voltage metal-clad switchgear in a sectionalized main-tie-main arrangement. The planning basis assumes three principal bus sections (A, B, and C), vacuum circuit breakers, feeder positions for data halls and emergency plants, capacitor-bank feeders, and auxiliary / station-service positions.

Planning item	Assumed value
Product family	Siemens NXAIR M
Voltage class	15 kV class
Main arrangement	Main-tie-main; sectionalized buses A, B, and C
Nominal bus basis	3000 A planning basis
Interrupting duty	40 kA planning basis
Use in package	Main campus 13.8 kV distribution switchgear at the customer substation

3. Standby Generation Platform

The emergency standby basis uses Rolls-Royce mtu 16V4000 DS2500 generator sets. The package assumes sixty units total, arranged as six isolated emergency generation plants with ten units assigned to each hall block. Each plant supports its associated hall emergency and essential-support loads. No merchant export or routine parallel operation across the point of interconnection is proposed.

Planning item	Assumed value
Product family	Rolls-Royce mtu 16V4000 DS2500
Per-unit electrical basis	2,500 kWe standby
Nominal voltage basis	13.8 kV alternator assumed for planning
Campus quantity	60 units total
Typical assignment	10 generator sets per hall emergency plant
Controls basis	Paralleling switchgear with export blocking, synchronizing control, and load sequencing

4. UPS Platform

The critical power basis uses Vertiv Liebert EXL S1 UPS modules rated at 1,200 kW and 480 V. The package assumes a 2N topology for each hall, with independent A and B paths and thirty-six total modules per hall (eighteen modules on each path).

Planning item	Assumed value
Product family	Vertiv Liebert EXL S1
Per-module basis	1,200 kW at 480 V
Topology basis	2N hall architecture; independent A and B power paths
Campus quantity	216 modules total
Per-hall basis	36 modules total; 18 per path

5. Capacitor Banks and Support Systems

The planning basis includes six switched capacitor-bank steps rated 6 Mvar each on the 13.8 kV campus system. The package also assumes dual 125 Vdc station battery systems, station service transformers, and hall service transformers required for the principal electrical arrangement.

These values are included to support engineering review of the interconnection concept only. Final manufacturer data sheets, certified dimensions, impedance data, and detailed control narratives will be provided during later design phases.